

Bond Convenience Yield and Convenience Features in Fixed-Income Markets

Executive summary

“Bond convenience” is best understood as the set of **non-cash-flow benefits** (“convenience features”) that some bonds provide to investors and intermediaries—benefits that can make investors willing to accept **lower yields (higher prices)** than would be implied by cash flows and default risk alone. In modern markets these benefits are tightly linked to **liquidity, collateral usefulness (repo/derivatives margin/clearing), regulatory treatment (HQLA and capital rules), settlement infrastructure, and tax treatment**. Empirically, these features generate a measurable **convenience yield (or convenience premium)**: a yield reduction relative to a comparable but less “convenient” asset. ¹

Key takeaways:

Convenience yield is not a single universal number. It is a **state-dependent wedge** between yields on “convenient” bonds (often sovereign benchmarks) and yields on alternative instruments that are similar in duration and (sometimes) credit risk but deliver fewer convenience services. The Federal Reserve has explicitly described the Treasury convenience yield as a wedge driven by **supply/demand for safe and liquid assets** and influenced by **regulation and investor demand**, and has linked time variation in this wedge to macro outcomes such as estimates of the natural real rate (r^*). ²

Official legal and regulatory definitions matter because they determine which bonds qualify for key convenience channels: - **U.S. “government securities”** are defined (for the Securities Exchange Act) as securities that are direct obligations of, or guaranteed as to principal or interest by, the United States. ³

- **U.S. “municipal securities”** are defined as securities that are direct obligations of, or guaranteed by, states/political subdivisions (and related entities), with additional statutory references to tax-exempt bond categories. ⁴

- **Tax status** for U.S. munis is anchored in the Internal Revenue Code’s general rule that qualifying state/local bond interest is excluded from federal gross income. ⁵

Across major markets: - **U.S. Treasuries:** multiple measures show sizable convenience yields and meaningful time variation. A St. Louis Fed analysis (2012–2025) shows, for example, an AAA–UST spread averaging **67 bp in 2012–2019** and **36 bp in 2025**, alongside maturity-dependent declines in swap and box-spread measures. ⁶

- **German Bunds:** IMF work measures a “Bund premium” (a relative convenience yield) that turns positive post-crisis and can reach several **tens of basis points**, increasing with scarcity and stress. ⁷

- **Japanese JGBs:** Bank of Japan research describes a repo market where “special collateral” repo rates can be **far below** general collateral rates (a direct manifestation of collateral convenience); the BOJ also operates a Securities Lending Facility aimed at supporting JGB market functioning amid scarcity. ⁸

- **Corporates and munis:** convenience is typically weaker and more investor-segment-specific (e.g., tax

preference for munis), but liquidity and regulatory eligibility can still produce measurable premia/discounts.

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Investor implications: convenience features reshape expected returns, hedging behavior, and relative value. They also create structural dislocations (e.g., repo “specialness,” on-the-run premia, swap spread compression, Treasury vs synthetic risk-free rates) that matter for portfolio construction, funding, and alpha strategies—especially under regulatory and balance-sheet constraints. 10

Assumptions used throughout (because not specified): market conditions are “generic” (not tied to a single stress episode), investor domicile and tax status are case-by-case, and “convenience yield” is treated as a **pricing wedge** attributable to convenience features after controlling (as best as available in the cited sources) for credit risk and duration. Where measures differ (AAA-UST vs swaps vs box spreads vs CIP-based wedges), the report treats them as **distinct empirical proxies** for overlapping components of convenience. 11

Concepts, definitions, and official legal anchors

Convenience yield in finance and its translation to bonds

In classical commodities finance, a “convenience yield” is the implicit benefit of holding the physical commodity (inventory), appearing as a term in the **cost-of-carry/forward pricing** relation and rationalizing persistent deviations from simple storage-cost arbitrage. The inventory-based concept is formalized in the theory of storage (Kaldor/Working/Brennan traditions) and is widely operationalized in empirical commodity futures studies. 12

In fixed income, the analogy is: certain bonds provide **convenience services** (liquidity, collateral, regulatory utility, safe-haven hedging), so investors accept **lower yields** relative to otherwise comparable instruments. A Federal Reserve note characterizes this explicitly as “a wedge between the interest rate of privately issued securities ... and the yield of government issued safe assets,” calling the wedge “the convenience yield of Treasury securities,” driven by supply/demand, issuance, regulation, and foreign/domestic demand. 2

A practical definition consistent with the official and academic sources used in this report:

Bond convenience yield (empirical): the yield difference between a “less convenient” benchmark (e.g., Aaa corporate yields, swap-implied rates, option-implied risk-free rates) and a “more convenient” bond (typically a sovereign benchmark), after matching maturity/tenor and attempting to control for credit risk and other frictions. 11

Convenience features of bonds

Below are the main convenience channels emphasized in official infrastructure/regulatory texts and in the empirical literature:

Liquidity and market depth. High turnover, tight bid-ask spreads, and large dealer/interdealer capacity reduce transaction costs and make the bond a preferred store of value and hedging instrument. Convenience can vary even within a single issuer’s curve (e.g., on-the-run vs off-the-run). Empirical Treasury

work shows that observable quoted bid-ask measures can misstate trading costs in off-the-run markets and that benchmark on-the-run conditions affect off-the-run liquidity. ¹³

Safety and safe-haven hedging. Bonds perceived as default-free in nominal terms (in their own currency) can provide hedging value in “bad times,” lowering required yields. Classic evidence shows Treasuries command a sizeable safety/liquidity-related yield reduction over long samples. ¹⁴

Regulatory treatment and balance-sheet utility. Liquidity regulation (LCR/HQLA) codifies which assets are treated as highly liquid—creating a structural demand for certain sovereigns/agencies and, to lesser extent, high-grade corporates/munis: - In the U.S. Federal Reserve’s liquidity rule, **Level 1 liquid assets** include Treasury securities and certain sovereign/official-sector securities meeting additional criteria, while **Level 2B** can include investment-grade corporate debt and even **investment-grade municipal obligations**. ¹⁵

- Basel LCR documentation states that HQLA are intended to be “converted easily and immediately in private markets into cash,” and it enumerates **Level 1 assets** that include central bank reserves and marketable claims on/guaranteed by sovereigns and central banks (subject to conditions including deep repo/cash markets and proven liquidity under stress). ¹⁶

Collateral usability (repo, central bank operations, derivatives margin). A major convenience channel is whether a bond is acceptable collateral, with low haircuts, across: - central bank credit/operations frameworks (ECB eligibility frameworks and BOJ eligible collateral guidance), ¹⁷

- secured funding markets (repo), including the ability of certain collateral to become “special,” generating financing benefits that get capitalized into bond prices, ¹⁸

- and derivatives margin frameworks (regulators and ISDA documentation), where eligible government debt is explicitly recognized and haircut schedules are prescribed. ¹⁹

Settlement and operational “moneyness.” Bonds settled and held in robust core infrastructures can be operationally preferred. For example: - The **Fedwire Securities Service** provides issuance/transfer/settlement services for marketable U.S. Treasury securities and is used to pledge collateral and facilitate repo transactions; transfers are described as final and irrevocable in Federal Reserve Financial Services materials. ²⁰

- The **DTCC/DTC settlement service** supports settlement for equity, corporate debt, and municipal debt securities (a different infrastructure channel than Fedwire Treasuries). ²¹

Tax treatment. Tax differences are a direct and legally grounded convenience feature: - U.S. Treasury marketable security earnings are subject to federal tax but exempt from state and local income taxes (TreasuryDirect; IRS Topic 403). ²²

- U.S. municipal bond interest is generally excluded from federal gross income under IRC §103 (subject to many conditions/exceptions). ⁵

Legal/official definitions that structure “convenience”

Because convenience often hinges on eligibility/definitions, these statutory anchors are especially relevant:

Under the U.S. Securities Exchange Act definition section: - “**Government securities**” are securities that are direct obligations of, or obligations guaranteed as to principal or interest by, the United States. ³

- “**Municipal securities**” are securities that are direct obligations of, or obligations guaranteed by, a state/

political subdivision and related entities (with additional references tied to tax-exempt industrial development bond concepts). ⁴

Under the U.S. Bankruptcy Code, repo-related definitions create legal convenience for certain collateral classes: - **“Repurchase agreement”** is defined to include repos on securities that are direct obligations of, or fully guaranteed by, the United States (and also “qualified foreign government securities,” among other instruments). ²³

- The Code provides protections for exercising contractual rights to liquidate/terminate/accelerate a repurchase agreement. ²⁴

These legal definitions effectively support the institutional role and collateral acceptability of U.S. sovereign debt (and certain foreign sovereign debt) in secured funding markets—one of the central “convenience” channels emphasized in asset-pricing research. ²⁵

Pricing mechanics and models of convenience in bond markets

A unifying “service flow” representation

Many models treat convenience as a **service flow** (liquidity/safety/collateral services) that accrues to the holder in addition to cash flows. A simple reduced-form decomposition of an observed bond yield can be written as:

$$y \approx r^{\text{frictionless}} + \text{credit term} + \text{liquidity/market friction term} - \text{convenience yield}.$$

This aligns with: - repo-based models where expected “specialness” reduces effective financing costs and raises cash bond prices; Duffie describes the GC–special repo spread as effectively an “additional dividend yield” that gets capitalized into the bond price. ²⁶

- macro-finance frameworks where Treasuries carry a premium relative to private borrowing rates; the Federal Reserve note explicitly generalizes a natural-rate model by introducing a wedge between private borrowing costs and yields of government safe assets (the convenience yield). ²

Key model families in the literature

Preferred habitat / segmented markets with limited arbitrage. A wide body of research explains term premia and relative bond pricing through investor clienteles and limits to arbitrage. IMF work on Bund scarcity explicitly uses a model intuition with preferred-habitat investors and risk-averse arbitrageurs to motivate why relative supply affects the “Bund premium,” especially at longer maturities and in high-volatility states. ²⁷

Safe-asset demand and “money-like” Treasuries. Krishnamurthy & Vissing-Jorgensen document that Treasury supply has large effects on yields/spreads and conclude that Treasuries’ safety and liquidity attributes reduce Treasury yields substantially over long horizons; they interpret Treasuries as “money-like,” meaning they deliver non-pecuniary services that investors value. ¹⁴

Repo specialness and collateral equilibrium. In collateral-based pricing, “special” securities command premia because they deliver financing benefits (low repo rates for borrowing cash against them, or the ability to cover short positions cheaply). Duffie’s repo model is explicit: specialness depends on shorting

demand and constraints on supplying the security into repo, and expected specialness increases the equilibrium price. ¹⁸

International convenience and CIP wedges. Du, Im, and Schreger measure the **relative convenience yield** of U.S. Treasuries versus other developed sovereigns using deviations from covered interest parity (CIP) between government bond yields (a “U.S. Treasury Premium”). They report that the five-year premium averaged about **21 bp pre-crisis**, rose to about **90 bp during the crisis**, and averaged about **–8 bp post-crisis**—interpreting this as a decline in Treasury specialness at medium/long maturities. ²⁸

Inference of “risk-free” rates from risky assets. Van Binsbergen, Diamond, and Grotteria infer risk-free rates from option prices to avoid contamination by convenience yields on safe assets, producing a term structure of convenience yields; their NBER summary states that Treasury convenience yield is about **40 bp** on average, larger below three-month maturity, and can **quadruple during the financial crisis**. ²⁹

Stress episodes and “inconvenience yields.” During severe dealer-balance-sheet constraints, Treasuries can temporarily trade *cheap* relative to other risk-free benchmarks; He, Nagel, and Song model and document “Treasury inconvenience yields” during the COVID-19 crisis, measured via spreads such as Treasury–OIS and repo intermediations. ³⁰

How convenience shapes the term structure and risk premia

Convenience is typically **maturity-dependent** because different maturities serve different roles: - Short bills are closest to “money substitutes” and collateral for overnight funding; measures tied to repo and near-money demand often emphasize the short end. ³¹

- Longer-term sovereign bonds serve as benchmarks for hedging, derivatives discounting spreads, and regulatory portfolios; empirical work (e.g., “Convenience Lost”) emphasizes that long-term convenience can be more supply-sensitive than short-term convenience. ³²

In a currency union (Eurozone), convenience yields can also act as a **shock absorber** when nominal short rates are common: ECB-hosted research materials argue that convenience yields explain a substantial share of variation in Eurozone yield differentials and respond to fiscal news. ³³

Evidence and case studies across bond classes

U.S. Treasuries

Core empirical proxies. A Federal Reserve Bank of St. Louis analysis summarizes three operational measures:

- **AAA–UST spread** (corporate AAA option-adjusted spread vs UST yields),
- **swap spreads** (swap rates minus Treasury yields),
- **box spreads** (option-implied “risk-free” rates minus Treasury yields).

They define Treasury convenience yield as the extra value investors place on Treasuries beyond discounted cash flows—reflecting liquidity and safety—and report a notable decline in recent years, especially at longer maturities. ⁶

Selected measured magnitudes (basis points). From the St. Louis Fed table of annual averages (all in bp):

- AAA-UST: **67 (2012-19) → 36 (2025)**;
- 10-year swap spread: **2 (2012-19) → -26 (2025)**;
- 18-month box spread: **34 (2012-19) → 9 (2025)**. ⁶

A Federal Reserve note uses the Aaa corporate minus 10-year Treasury spread as a convenience yield measure (following Krishnamurthy & Vissing-Jorgensen) and links sustained variation to macro estimates of r^* via a wedge between private borrowing costs and safe-asset yields. ²

Long-run estimates of Treasury “specialness.” Krishnamurthy & Vissing-Jorgensen report that Treasury yields are reduced by about **73 bp on average (1926-2008)**, driven by both liquidity and safety attributes.

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Collateral channel via repo “specialness.” Repo specialness is a direct, transaction-level convenience yield: Duffie defines specialness as the difference between the general collateral repo rate and the specific collateral rate, and argues this difference is effectively an additional dividend yield capitalized into the cash price. ²⁶

German Bunds

IMF research (“Bund premium”) measures the convenience-yield advantage of Bunds relative to other sovereign safe assets after correcting for FX-swap frictions (CIP), sovereign credit risk differentials, and liquidity differences. It finds Bund specialness increased after the global financial crisis and spiked during the euro-area debt crisis, remaining positive thereafter. ³⁵

From the IMF figures (visual evidence): - Bund premia vs the U.S. are **negative pre-crisis** but become **positive post-crisis**, reaching roughly **0.3%-0.5%** ($\approx 30-50$ bp) in the mid-to-late 2010s, with larger spikes around major stress episodes. ⁷

Mechanism and policy link: the IMF paper explicitly points to Bund scarcity (fiscal stance plus ECB QE reducing free float) and shows a negative relationship between relative German debt supply and the premium, steeper at longer maturities and when bond-market volatility is high. ²⁷

Japanese JGBs

Collateral and repo structure. Bank of Japan (IMES) research distinguishes: - **General collateral (GC) repo** to borrow/lend funds, priced close to risk-free rates; and

- **Special collateral (SC) repo** to borrow/lend specific securities, often priced far below GC rates (i.e., large “specialness” when a particular issue is scarce/valuable as collateral). ³⁶

That SC-GC wedge is conceptually identical to Duffie’s “specialness” channel and constitutes a convenience yield embedded in the cash prices of the “special” bonds. ³⁷

Central bank mitigation of scarcity. The BOJ explicitly operates a **Securities Lending Facility (SLF)** to provide a temporary/secondary source of JGBs to market participants with the stated purpose of improving liquidity and maintaining smooth market functioning. ³⁸

Quantitative easing and scarcity effects. IMF research on Japan finds that BOJ JGB purchases can generate security-level scarcity effects that change market liquidity (a key driver of convenience/illiquidity premia). ³⁹

High-grade corporate bonds

High-grade corporates can be “convenient” relative to lower-grade credit but are typically **less convenient than benchmark sovereigns** because they provide weaker safety, poorer repo specialness characteristics, and less favorable regulatory treatment (often with higher haircuts and caps).

Empirical work by Bao, Pan, and Wang finds that corporate bond illiquidity is a major driver of aggregate credit spreads for high-rated bonds; in their analysis, an aggregate illiquidity measure explains a large fraction of monthly changes in high-grade spreads (reported R-squared ranges about **47%–60%** for AAA through A). ⁴⁰

Regulatory channel: under the U.S. LCR rule, **investment-grade corporate debt securities** can qualify as **Level 2B liquid assets** (subject to conditions and caps), which can create an institutional demand component for the most eligible corporate paper—but it remains less privileged than Treasuries as Level 1 assets. ¹⁵

Municipal bonds

Municipal bonds have a unique and legally anchored convenience channel through **tax preference**. Under IRC §103, qualifying state and local bond interest is excluded from federal gross income, lowering required pre-tax yields for taxable investors. ⁵

Empirical evidence supports strong tax sensitivity: Ang, Bhansali, and Xing identify tax effects using taxable “market discount” municipal bonds and report yields about **25 bp higher** than comparable municipal bonds not subject to those taxes, underscoring how tax treatment is capitalized into muni pricing. ⁴¹

Municipals also exhibit microstructure-driven illiquidity: research on municipal bond trading emphasizes opacity and decentralized dealer markets, with meaningful trading frictions and price dispersion—reducing liquidity convenience relative to Treasuries even as tax convenience raises demand. ⁴²

Regulatory channel: under the U.S. LCR rule, an **investment-grade municipal obligation** can qualify as a **Level 2B liquid asset**, which is a notable “regulatory convenience” feature for banks (again subject to limits and haircuts). ¹⁵

Comparison across bond types

Comparative table of convenience features and typical premia

The table below summarizes typical patterns. “Typical convenience yield” is reported as an **order-of-magnitude** range supported by the cited empirical measures; it should be interpreted as **time-varying and state-dependent**, not a constant. ⁴³

Bond type (illustrative examples)	Liquidity (secondary market)	Repo eligibility / “specialness” potential	Regulatory convenience (HQLA/ capital)	Tax convenience (typical for taxable investor)	Collateral beyond repo (CB ops / derivatives IM)	Typical convenience yield proxy (indicative)
Reserve- currency sovereign (U.S. Treasures)	Very high; benchmark pricing and hedging instrument ⁴⁴	Core repo collateral; on-the-run issues can go “special,” embedding financing value ¹⁸	Level 1 HQLA in U.S. LCR; central bank eligible (not sufficient alone but commonly required) ⁴⁵	Interest federally taxable but exempt from state/ local income tax ²²	Eligible collateral under derivatives margin schedules and central bank frameworks ⁴⁶	~30–70 bp in recent spread-based measures; long-run estimate ~73 bp (1926– 2008) ⁴⁷
Euro “safe- haven” sovereign (German Bunds)	High; euro benchmark	High repo collateral value; scarcity episodes increase specialness/ convenience ⁴⁸	Eurosystem- eligible collateral framework; scarcity affected by QE and issuance ⁴⁹	Tax depends on investor domicile; not a universal feature	Broad collateral usability in Eurosystem operations (eligibility database & criteria) ⁵⁰	IMF “Bund premium” often in the tens of bp post-crisis (≈0.3–0.5%) vs some comparators ⁷
High- savings sovereign with active collateral markets (Japanese JGBs)	High domestically; liquidity can be impaired by scarcity	Repo market explicitly distinguishes GC vs special collateral; SC rates often far below GC (specialness) ³⁶	Central bank eligible collateral; BOJ provides securities lending to support functioning ⁵¹	Depends on investor type; Japan has detailed tax regimes/ exemptions ⁵²	BOJ eligible collateral + clearinghouse collateral policies (e.g., JSCC accepts JGBs) ⁵³	Often evidenced via repo specialness (GC–SC wedge); magnitude varies by issue and scarcity (directionally “can be large”) ⁵⁴

Bond type (illustrative examples)	Liquidity (secondary market)	Repo eligibility / “specialness” potential	Regulatory convenience (HQLA/ capital)	Tax convenience (typical for taxable investor)	Collateral beyond repo (CB ops / derivatives IM)	Typical convenience yield proxy (indicative)
U.S. municipal bonds	Generally lower than Treasury; fragmented issues and opacity ⁴²	Repo eligibility varies; not the universal repo benchmark	Can qualify as Level 2B HQLA (investment grade) for U.S. banks ¹⁵	Federal tax exemption under IRC §103 (subject to rules), often central to pricing ⁵⁵	Not typical “top-tier” derivative margin collateral; eligibility more limited than sovereigns	Tax-driven yield effects can be tens of bp ; e.g., taxed muni structures ~25 bp higher yields in identified settings ⁵⁶
High- grade corporates (e.g., AAA/ AA)	Moderate; episodic liquidity stress; dealer balance- sheet effects	Repo possible but with higher haircuts and less “specialness” than benchmarks	Can qualify as Level 2B HQLA (investment grade; conditions apply) ¹⁵	Typically fully taxable	Eligible collateral in derivatives/ margin frameworks, often with higher haircuts than sovereigns ¹⁹	“Convenience” usually shows up as liquidity premium/ illiquidity spread ; illiquidity explains large share of high- grade spread variation ⁵⁷

Mermaid flow diagram of how features create convenience yield

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flowchart TD
    A[Bond instrument] --> B[Cash flows<br/>(coupons, principal)]
    A --> C[Convenience features]
    C --> C1[Liquidity<br/>(depth, bid-ask, turnover)]
    C --> C2[Collateral usability<br/>(repo, CCPs, margin)]
    C --> C3[Regulatory treatment<br/>(HQLA, capital, leverage)]
    C --> C4[Settlement & ops<br/>(CSD, finality, fails regime)]
    C --> C5[Tax treatment<br/>(exemptions, withholding)]
    C1 --> D[Non-pecuniary service value]
    C2 --> D
    C3 --> D
    C4 --> D
    C5 --> D
    B --> E[Fundamental value]
    D --> F[Convenience premium<br/>(higher price)]
  
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E --> G[Observed price]
F --> G
G --> H[Observed yield\nlower when premium higher]

```

This diagram matches the conceptual structure used in official and academic sources: convenience features produce a valued service flow; the bond's price reflects both discounted cash flows and the priced service value, reducing observed yields. ⁵⁸

Charts and empirical illustrations

Treasury convenience yield measures over time (annual averages)

The St. Louis Fed table provides annual averages for several convenience-yield proxies. Below is a compact visualization using the published values (bp). ⁶

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xychart-beta
  title "U.S. Treasury convenience yield proxies (bp, annual averages)"
  x-axis ["2012-19", "2020", "2021", "2022", "2023", "2024", "2025"]
  y-axis "bp" -40 --> 90
  line "AAA-UST" [67,79,50,63,51,37,36]
  line "10Y swap spread" [2,-1,1,1,-6,-18,-26]
  line "18M box spread" [34,30,51,36,-17,-2,9]

```

Interpretation (within the limitations of each proxy): - Declining AAA-UST and box spreads suggest reduced Treasury “convenience” in recent years, consistent with increased Treasury supply and changes in the relative value of long-maturity Treasuries. ⁶

- Negative long-tenor swap spreads (e.g., -26 bp at 10Y in 2025) imply that the Treasury yield can sit above the matched swap rate, which can be interpreted as reduced Treasury convenience or increased swap convenience/constraints depending on market structure—hence the importance of triangulating across measures. ⁵⁹

Investor implications and open research questions

Implications for investors, issuers, and risk managers

Relative value is “convenience-sensitive.” Comparing sovereigns to swaps, agencies, or corporates is not just about default risk and duration; it is also about convenience channels that can flip sign in stress (e.g., “inconvenience yields” when balance sheets are constrained). This is central to Treasury-OIS dynamics in episodes like COVID-19 market stress. ⁶⁰

Funding and collateral strategy is a return driver. Repo specialness can turn a bond position into a “yield + financing benefit” package; conversely, losing specialness (or facing higher haircuts) can erase apparent carry. Duffie’s framing of specialness as an extra dividend that is priced into the cash bond is directly applicable to practical relative value and hedging decisions. ³⁷

Regulatory demand is structural. When regulations define HQLA categories (Treasuries as Level 1; high-grade corporates/munis as Level 2B with haircuts/caps), they create predictable investor clienteles and supply-demand elasticities that affect term premia and cross-asset spreads. ⁶¹

Tax-aware portfolio construction is essential in municipals. For taxable investors, muni valuation hinges on the legal tax exemption and on market microstructure/illiquidity; empirical work shows tax effects are capitalized into yields, but trading frictions and investor segmentation create deviations from simple tax-arbitrage predictions. ⁶²

International investors face “currency-hedged convenience.” CIP-based measures show that relative convenience yields can be studied only after currency hedging; the “U.S. Treasury Premium” approach operationalizes this and shows that medium/long Treasury specialness can diminish over time, even when short-dated convenience remains. ⁶³

Open research questions

Decomposition and identification. How should one robustly decompose observed sovereign yields into (i) frictionless risk-free rate, (ii) default premium (if any), (iii) liquidity premium, and (iv) collateral/regulatory convenience—especially when proxies disagree (AAA-UST vs swap spreads vs box spreads)? The Federal Reserve and St. Louis Fed discussions highlight the practical importance of this measurement problem. ⁶⁴

Endogenous convenience under regulation and dealer constraints. Recent work links diminishing Treasury convenience to intermediary balance-sheet constraints and auction absorption capacity, and models stress periods where Treasuries can become “inconvenient.” This raises questions about how post-crisis regulation changes the equilibrium convenience yield across maturities and states of the world. ⁶⁵

Scarcity management by central banks. ECB/BOJ collateral frameworks, QE programs, and securities lending facilities (e.g., BOJ SLF) all interact with convenience by changing free float and collateral availability. Quantifying the welfare and market-function tradeoffs remains an active area. ⁶⁶

Convenience spillovers across sovereign issuers. Euro-area work (including IMF and later research presentations) emphasizes cross-country spillovers of debt supply into convenience yields; understanding substitution patterns across “safe” vs “risky” sovereigns is a frontier for term-structure and fiscal-monetary interaction research. ⁶⁷

¹ ⁶ ¹¹ ⁴³ ⁴⁴ ⁴⁷ ⁵⁹ Are U.S. Treasuries Still 'Convenient'? | St. Louis Fed

<https://www.stlouisfed.org/on-the-economy/2025/oct/are-u-s-treasuries-still-convenient>

² ⁵⁸ ⁶⁴ The Fed - Convenience Yield as a Driver of r^*

<https://www.federalreserve.gov/econres/notes/feds-notes/convenience-yield-as-a-driver-of-r-20240903.html>

³ ⁴ 15 U.S. Code § 78c - Definitions and application | U.S. Code | US Law | LII / Legal Information Institute

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⁵ ⁵⁵ ⁶² 26 U.S. Code § 103 - Interest on State and local bonds

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